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Equine Cardiology

Effect of N-Butylscopolammonium Bromide and Metamizol Sodium on Heart Rate, Blood Pressure, and Echocardiographic Measurements in Warmblood Horses With Aortic and Mitral Valve Regurgitation

Alexander Dufourni | Marie Demeyere | Ingrid Vernemmen | Glenn Van Steenkiste | Lize-Maria Verhaeghe | Gunther van Loon | Annelies Decloedt

Equine Cardioteam Ghent University, Department of Internal Medicine, Reproduction and Population Medicine, Faculty of Veterinary Medicine, Ghent University, Merelbeke, Belgium

Correspondence: Alexander Dufourni (alexander.dufourni@ugent.be)

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ABSTRACT

Background: Administration of N-butylscopolammonium bromide (NBB) with metamizol sodium (NBBM) causes tachycardia and hypertension, and has been associated with accentuated cardiac murmurs.

Objectives: To evaluate the effect of a pharmacological challenge using NBBM on valvular regurgitation and cardiac murmurs in horses with aortic (AR) and mitral valve regurgitation (MR).

Animals: Twenty Warmblood horses with AR and 20 with MR.

Methods: Cardiac auscultation and two-dimensional (2D), M-mode, color flow, pulsed wave Doppler flow, and tissue Doppler echocardiography were performed with simultaneous ECG recording and non-invasive blood pressure measurements during rest and pharmacological challenge using 0.2 mg/kg NBB and 25 mg/kg metamizol sodium. Cardiac dimensions and functional parameters were assessed. Regurgitant jet size was subjectively evaluated, and jet area was measured. Data at rest and during pharmacological challenge were compared using repeated measures analysis or related samples Wilcoxon signed rank test. *p*-values were adjusted using the Benjamini-Hochberg method.

Abbreviations: 2D, two-dimensional; A_m, peak left ventricular radial wall motion velocity during late diastole; Ao, aorta; $AoDd_{end}$, aortic diameter at sinotubular junction at end-diastole; $AoDs_{peak}$SAX, aortic short axis diameter at peak-systole; $AoDs_{peak}$, aortic diameter at sinotubular junction at peak-systole; AoV, aortic valve; AR, aortic regurgitation; BHa, Benjamini-Hochberg adjusted P-value; BP, blood pressure; ch, chordal level; CI, confidence interval; CO, cardiac output; DAP, diastolic arterial blood pressure; EF, ejection fraction; E_m, peak left ventricular radial wall motion velocity during early diastole; ET, ejection time; FAC, fractional area change; FS, fractional shortening; HR, heart rate; IQR, interquartile range; $IVSd_{end}$, interventricular septal thickness at end-diastole; $IVSs_{peak}$, interventricular septal thickness at peak-systole; L, left; LA FAC, left atrial fractional area change; LA, left atrium/atrial; LAAa, left atrial area at maximal atrial contraction; $LAAd_{end}$, left atrial area at end-diastole; LAAp, left atrial area at the onset of the P wave; $LAAs_{end}$, left atrial area at end-systole; $LAD_{max}s_{end}$, maximal left atrial diameter at end-systole measured parallel to the mitral valve annulus; $LADs_{end}$SAX, short axis diameter of the left atrium at end-systole; $LADs_{end}$, left atrial diameter at end-systole measured parallel to the mitral valve annulus; LAX, long axis view; LV FAC, left ventricular fractional area change; LV FS, left ventricular fractional shortening; LV, left ventricle/ventricular; LVA, left ventricular area; $LVFWd_{end}$, LV free wall thickness at end-diastole; $LVFWs_{peak}$, LV free wall thickness at peak-systole; $LVIDd_{end}$, LV internal diameter at end-diastole; $LVIDs_{peak}$, LV internal diameter at peak-systole; LVOT, left ventricular outflow tract; $LVVd_{end}$, LV volume at end-diastole; $LVVs_{end}$, LV volume at end-systole; MAP, mean arterial blood pressure; ModSi, Simpson's modified method of disks; NBB, N-butylscopolammonium bromide; NBBM, N-butylscopolammonium bromide and metamizol sodium; NIBP, non-invasively measured blood pressure; PA, pulmonary artery; $PADd_{end}$, PA diameter at end-diastole; $PADs_{peak}$, PA diameter at peak-systole; PEP, pre-ejection period; PW TDI, pulsed wave tissue Doppler imaging; PWD, pulsed wave Doppler; R, right; RV, right ventricle; $RVIDd_{end}$, RV internal diameter at end-diastole; $RVIDs_{peak}$, RV internal diameter at peak-systole; $RWTd_{end}$, relative wall thickness at end-diastole; SAP, systolic arterial blood pressure; SAX, short axis view; SD, standard deviation; S_m, peak left ventricular radial wall motion velocity during peak-systole; SV, stroke volume; TDI, tissue Doppler imaging; Vel_{max}, maximum velocity; VTI, velocity time integral.

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Results: Compared with rest, pharmacological challenge resulted in decreased atrial and ventricular dimensions and increased aortic and pulmonary dimensions ($P_{BHa} \leq 0.02$). Left ventricular end-diastolic diameter decreased in horses with AR ($P_{BHa} < 0.02$) but not in horses with MR. Pharmacological challenge resulted in increased median cardiac murmur intensity ($P_{BHa} = 0.03$ and $P_{BHa} < 0.02$) and jet area ($P_{BHa} \leq 0.02$ and $P_{BHa} = 0.03$) in horses with AR and MR, respectively.

Conclusions and Clinical Importance: Pharmacological challenge with NBBM affects cardiac chamber sizes, regurgitant jet, and cardiac murmur intensity in horses with AR and MR. Influences on heart rate and blood pressure should be considered during longitudinal follow-up examinations.

1 | Introduction

Cardiac murmurs arise from high-velocity turbulent blood flow or vibrating cardiac structures. Aortic regurgitation (AR) and mitral valve regurgitation (MR) are common valvular disorders causing cardiac murmurs in horses [1, 2]. Characterization of a cardiac murmur based on its location, timing, and intensity provides valuable information on the possible underlying cause. Therefore, physiological flow murmurs need to be distinguished from pathological murmurs indicating structural heart disease [1, 3–5].

In human patients, physiological (post-exercise) and pharmacological stress testing can modify pre-existing cardiac murmurs or identify transient or intermittent cardiac murmurs that are absent under resting conditions [6–8]. The persistence and grade of severity of valvular regurgitation during cardiac stress testing have been described in human patients as indicators of disease severity and allow the identification of individuals at risk where valvular insufficiency might become clinically relevant [6, 9–11]. In Standardbred trotters, the presence of increased tricuspid valve regurgitation after exercise testing has been associated with a history of poor performance [7]. Data on pharmacological stress testing in horses with valvular regurgitation currently is lacking in the veterinary literature. However, as in human patients, the persistence or increase in valvular regurgitation during stress testing in horses might provide useful information about its current or future clinical importance. Because exercise echocardiography in horses is impractical and restricted to the immediate postexercise period [7, 12, 13], pharmacological stress testing might be applied and result in good image quality during the test [13].

A combination of N-butylscopolammonium bromide and metamizol sodium (NBBM) is a widely used drug combination in equine medicine for its antispasmodic, bronchodilatory, and analgesic properties with anticholinergic antimuscarinic parasympatholytic effects [14–16]. Transient tachycardia and hypertension have been reported to be the main cardiovascular effects of the drug [16, 17], and resulted in decreased left atrial (LA) and diastolic left ventricular (LV) dimensions as well as increased aortic and pulmonary dimensions in healthy Warmblood horses [18]. Metamizole sodium consists of two primary active metabolites, 4-methylaminoantipyrine and 4-aminoantipyrine, which exhibit analgesic, antipyretic, and spasmolytic properties, and do not significantly affect the patient's cardiovascular or hemodynamic function after a single bolus injection [19, 20]. Because NBBM has been reported to temporarily accentuate cardiac murmurs in horses [21], the drug might be useful to diagnose subclinical valvular regurgitations for which murmur intensity is low. Similar to the effect of NBBM, increased blood pressure (BP) caused by pain during an episode of gastrointestinal colic might explain why cardiac murmurs are more pronounced in some horses with colic. Horses exhibiting cardiac murmurs during NBBM challenge might need further echocardiographic examination for the assessment of potential valvular regurgitation. As such, NBBM might be useful as a diagnostic test in horses with low-intensity AR or MR murmurs.

Our aim was to evaluate the effects of NBBM on echocardiographic measurements and function, jet size, and audibility of the cardiac murmur in Warmblood horses with AR and MR. We hypothesized that pharmacological challenge would result in increased valvular regurgitation and murmur intensity.

2 | Materials and Methods

2.1 | Animals and Experimental Design

An exploratory pre-post study in horses with MR and AR was performed. Data collection occurred between November 2022 and January 2024. The study sample included 20 Warmblood horses with AR and 20 Warmblood horses with MR. The study was approved by the ethical committee of the Faculty of Veterinary Medicine and Bioscience Engineering (EC2020/100), with written owner informed consent obtained. Inclusion criteria were: Warmblood, regular heart rhythm at rest, and known AR or MR in the absence of clinical signs of heart failure. Cases were selected based on the owners' willingness to participate in the study until 20 horses per group were enrolled. The severity of valvular regurgitation was determined according to a newly developed classification scale, based on literature and in collaboration with the Equine Department of the University of Zurich. For AR, the severity of valvular regurgitation was classified as trivial, mild, moderate, or severe based on a scoring system including LV volume, jet size compared to the left ventricular outflow tract (LVOT) assessment area, and subjective LV size (Data S1). For MR, the LA area, jet size compared to the LA area, jet duration, and subjective LA size assessed on long axis (LAX) and short axis (SAX) views were included in the classification scale (Data S2). Horses were allocated to either the AR or MR group based on the most severe valvular regurgitation identified. Concurrent trivial or mild valvular regurgitations were allowed for inclusion in the study, provided that the severity grade of the concurrent regurgitation was equal to or less than that of the primary regurgitation. Height at the withers and body weight were recorded for all horses. Body condition score was recorded based on the Henneke body condition score scale [22]. All horses underwent cardiac auscultation and echocardiography at rest and during pharmacological challenge. An ECG using the recently proposed Delta Einthoven configuration was recorded

simultaneously using a Televet device (Engel Engineering Services GmbH, Heusenstamm, Germany) to determine heart rate (HR) and to monitor for pathological dysrhythmias [23]. First, a general clinical examination, resting cardiac auscultation, resting BP measurement, and resting echocardiographic examination were performed. After the resting examination, a standardized lunging exercise test was performed in the majority of horses to screen for exercise-associated dysrhythmias. The test included 5 min of walk, 10 min of trot, 4 min of canter, and 1 min of gallop, followed by 5 min of walk [24]. After the exercise test, a resting period of approximately 1 h was provided in a box with hay and fresh water. Horses that did not perform the exercise test underwent pharmacological challenge immediately after the resting examination. Pharmacological challenge was performed by IV administration of 0.2 mg/kg N-butylscopolammonium bromide and 25 mg/kg metamizol sodium (NBBM, Buscopan compositum, Boehringer Ingelheim Vetmedica GmbH, Germany). Echocardiographic image acquisition started within 5 min after IV drug administration and lasted for approximately 25 min, while the Televet ECG indicated sinus tachycardia.

2.2 | Cardiac Auscultation and Recording of Cardiac Sounds

Cardiac auscultation was performed using the 3M Littmann CORE digital stethoscope at rest before image acquisition, immediately after the gallop of the exercise test and within 2 min after IV drug administration during pharmacological challenge by one experienced operator, at the point of maximal cardiac murmur intensity. Additionally, at each auscultation time point, three 15-s loops of cardiac sounds were recorded from the left hemithorax for offline analysis. Recordings were transferred by Bluetooth to iOS and stored in a personalized Eko Health Inc. cloud. Heart rate was calculated from the recorded phonogram (paper speed 25 mm/s) by dividing 1500 by the distance (in mm) between two consecutive first heart sounds. The mean value of three cardiac cycles was used for further analysis. Cardiac murmurs were graded both during live auscultation and based on the recordings.

2.3 | Blood Pressure Measurements

Repetitive, consecutive, non-invasive blood pressure (NIBP) measurements were performed using an indirect oscillometric BP device (Cardell Veterinary Monitor, 9401, Midmark, Florida, USA), with the cuff placed around the base of the tail at the level of the coccygeal artery to determine diastolic (DAP), mean (MAP), and systolic (SAP) arterial blood pressure. The cuff width was 9 cm to achieve the recommended width-to-tail circumference ratio ranging between 0.4 and 0.6 [18, 25]. Blood pressure measurements were considered adequate for further analysis when the pulse rate obtained by the oscillometric device did not differ by more than 20% from the electrocardiographically-obtained HR [26]. Measurements were not corrected for differences in vertical distance between the coccygeal artery and the heart base [26]. At rest, five BP measurements were recorded before image acquisition. During pharmacological challenge, a minimum of 15 BP measurements were performed simultaneously with the acquisition of echocardiographic images at an approximate running time of 1 min per measurement. The mean value was used for further analysis.

2.4 | Echocardiography

Echocardiography was performed using a 1.5–4.6 MHz phased array probe (M5Sc-RS) at a frequency of 1.7/3.3 MHz with simultaneous ECG recording (base-apex; Vivid IQ, GE Healthcare, Diegem, Belgium) by one experienced operator. The examinations at rest and during pharmacological challenge included two-dimensional (2D) and M-mode views to assess cardiac structures, color Doppler images to assess valvular regurgitation, and pulsed wave Doppler (PWD) images to assess aortic flow velocities. Pulsed wave tissue Doppler imaging (PW TDI) was used to assess LV wall motion. Two-dimensional images included two right parasternal LAX views with either focus on LA or LV (R-LVLAX), a right parasternal LAX LVOT view (R-LVOT), a right parasternal SAX view of the aorta (Ao) and LA (R-AoSAX), and a left parasternal LAX view of the left ventricle, mitral valve, and left atrium (L-LVLAX). Two-dimensional and pulsed wave tissue Doppler images (PW TDI) were obtained from a right parasternal SAX view of the LV at chordal level (R-LV_{ch}SAX). Color flow Doppler imaging was performed to evaluate valvular regurgitation. Aortic valve regurgitation was assessed on R- and L-LVOT views, whereas MR was assessed on the L-LVLAX view with focus on the LA. Aortic blood flow velocity was evaluated by PWD on a L-LVOT view. Pulsed wave tissue Doppler imaging allowed assessment of the LV free wall velocity with a sample volume width of 5.0 mm and the PWD cursor positioned within the subendocardial myocardium of the LV at end-diastole. Image acquisition occurred in a predetermined order, starting with 2D parasternal views, followed by color flow, PWD flow, and PW TDI echocardiography. A minimum of three loops of three consecutive cardiac cycles were recorded for off-line analysis. Initiation of image acquisition during pharmacological challenge on either the left or right side was randomly assigned.

2.5 | Off-Line Analysis

The recorded auscultation files were retrospectively evaluated by three experienced independent observers. Cardiac murmurs were graded based on timing (no cardiac murmur, early-to-mid, mid-to-end, holo, pan) and intensity (grade 0–6/6 for live auscultation, grade 0–5/6 for recordings) [4]. Cardiac murmur timing was reported when at least two of the three observers agreed on the classification of timing. The murmur intensity during live auscultation and the median intensity of the recording assessed by three observers were reported at rest, immediately after exercise, and during pharmacological challenge. Televet ECGs were analyzed retrospectively to screen for pathological dysrhythmias or conduction abnormalities. Echocardiographic images were analyzed using EchoPAC Software Version 203 (GE Healthcare, Diegem, Belgium) by a single observer who was blinded to the respective horse. For each ultrasonographic variable, the mean or median of three cardiac cycles was used for further analysis. Cardiac cycles following physiological or pathological dysrhythmias were not used for further analysis. Mean HR was

determined from the RR intervals before the analyzed cardiac cycles. An overview of the echocardiographic measurements is presented in Data S3. Diameters, areas, and volumes were normalized to a body weight of 500 kg by allometric scaling [2, 27]:

- For diameters: measured diameter/$BW^{1/3} \times 500^{1/3}$
- For areas: measured area/$BW^{2/3} \times 500^{2/3}$
- For volumes: measured volume/BW × 500

Regurgitant blood flow was assessed by evaluating the measured jet area, a semiquantitative jet score, and subjectively assessed jet size before and after the administration of NBBM. First, the regurgitant jet area was measured as the most objective method for quantitatively assessing the regurgitant jet. Turbulent regurgitant blood flow was depicted by a green color originating from the closed valve at an aliasing velocity of 50–70 cm/s. Jet areas were measured for AR on a R- and L-LVOT views and for MR on a L-LVLAX view. The aortic regurgitant jet area was measured by tracing the green-colored jet circumference during mid-to-end diastole. In the case of MR, the cumulative jet area was assessed by measuring the largest respective jet area of each jet during systole. Second, a semiquantitative assessment was performed using a jet score, as defined in Data S1 for AR and Data S2 for MR. Third, the regurgitant jet size was subjectively assessed by evaluating the jet's visual appearance throughout the cardiac cycle. This jet size was compared between rest and during the NBBM challenge (not blinded), and classified as decreased, similar, or increased relative to rest.

2.6 | Data Analysis

Statistical analysis was performed using SPSS Statistics 29 (IBM, Brussels, Belgium). Normality of the distribution was evaluated by visual inspection of the QQ plots and histograms, using Kolmogorov–Smirnov and Shapiro–Wilk tests. Normally distributed data are presented as mean and SD. Non-normally distributed data are presented as median and interquartile range (Q1; Q3). Clinical variables were compared between horses with AR and MR using analysis of variance for normally distributed data and the Mann–Whitney U test for non-normally distributed data ($p < 0.05$ was considered significant). Mean HR, mean BPs as well as means or medians of echocardiographic variables were compared between rest and pharmacological challenge for horses with AR and MR. Normally distributed data were analyzed using a repeated measures analysis of variance with NBBM treatment as within-subject factor and valvular regurgitation (AR or MR) as between-subject factor. If pharmacological challenge resulted in a significantly different treatment effect in horses with MR compared with AR ($p < 0.05$), variables were further analyzed using a paired sample *t*-test. Non-normally distributed data were compared using a related samples Wilcoxon signed rank test to compare measurements before and after pharmacological challenge. The mean difference between rest and pharmacological challenge and its 95% confidence interval (CI) were determined for normally distributed data as a measure of effect size. Cardiac murmur timing was compared between rest and exercise and between rest and pharmacological challenge using a Pearson chi-square test. Cardiac murmur intensity was compared between rest and exercise and between rest and pharmacological challenge using a related samples Wilcoxon signed rank test. The *p*-values were adjusted post hoc using the Benjamini-Hochberg (BH) method to correct for a false discovery rate set to 5% (type I error: $\alpha = 0.05$). The critical value for each original *p*-value was calculated using the formula: (rank of the sorted original *p*-value/total number of *p* values) × false discovery rate. Adjusted *p*-values were calculated using the formula: original sorted *p*-value × (total of ranks/rank of the corresponding sorted *p*-value). The BH adjusted *p*-values (P_{BHa}) are reported along with the original *p*-values in the tables and P_{BHa} values < 0.05 were considered significant [28].

3 | Results

The study sample included 26 geldings, 12 mares, and two stallions. Based on the classification system, the study included one horse with trivial, seven horses with mild, five horses with moderate, and seven horses with severe AR, as well as eight horses with trivial, 11 horses with mild, and one horse with moderate MR. Mean ± SD body weight was 565 ± 56 kg. Median (Q1; Q3) body condition score was 5 (4; 6)/9. No significant differences were found between horses with AR and MR regarding sex, weight, and BCS. The median age of horses with AR, 21 (16; 24) years, differed significantly from horses with MR, 11 (8; 15) years ($P_{BHa} < 0.02$). Mean height at the withers was 167 ± 7 cm in horses with AR and 172 ± 6 cm in horses with MR ($P_{BHa} < 0.02$). A lunging exercise test was performed in 13/20 horses with AR and 15/20 horses with MR. Heart rate upon post-exercise auscultation was significantly higher compared with rest (80 ± 15 vs. 38 ± 6 beats per minute [bpm] for AR and 75 ± 14 vs. 38 ± 6 bpm for MR; $P_{BHa} < 0.02$, respectively). Pharmacological challenge with NBBM resulted in significantly higher HRs, DAPs, MAPs, and SAPs compared with rest (Figure 1). Left atrial dimensions decreased, with an increased LA FAC LAX ($P_{BHa} < 0.02$). End-diastolic LV area, volume, and internal diameter decreased on 2D- and M-mode views, with increased interventricular, free wall, and relative wall thickness on M-mode views. Left ventricular FAC LAX decreased as a result of a decreased LV end-diastolic area during pharmacological challenge ($P_{BHa} < 0.02$). Left ventricular SV and EF decreased, whereas CO increased ($P_{BHa} < 0.02$). Aortic and pulmonary diameters increased ($P_{BHa} < 0.02$).

In horses with AR, median right and left regurgitant jet areas increased after administration of NBBM compared with rest ($P_{BHa} \leq 0.02$; Table 1). The regurgitant jet score remained the same in 15/20 horses, whereas a higher jet score was assigned in 5/20 horses (Table 2). The subjectively assessed jet size increased during pharmacological challenge in 17/20 horses, whereas it appeared similar to rest in 3/20 horses.

For horses with MR, median cumulative left regurgitant jet area increased after administration of NBBM ($P_{BHa} \leq 0.03$; Table 1). The regurgitant jet score increased significantly after administration of NBBM in horses with MR ($P_{BHa} = 0.03$; Table 2) with higher MR jet scores assigned in 11/20 horses and unchanged jet scores in 9/20 horses compared to rest. The subjectively assessed jet size of the regurgitant jet increased during pharmacological challenge in 16/20 horses with MR, whereas it appeared similar to rest in 4/20 horses.

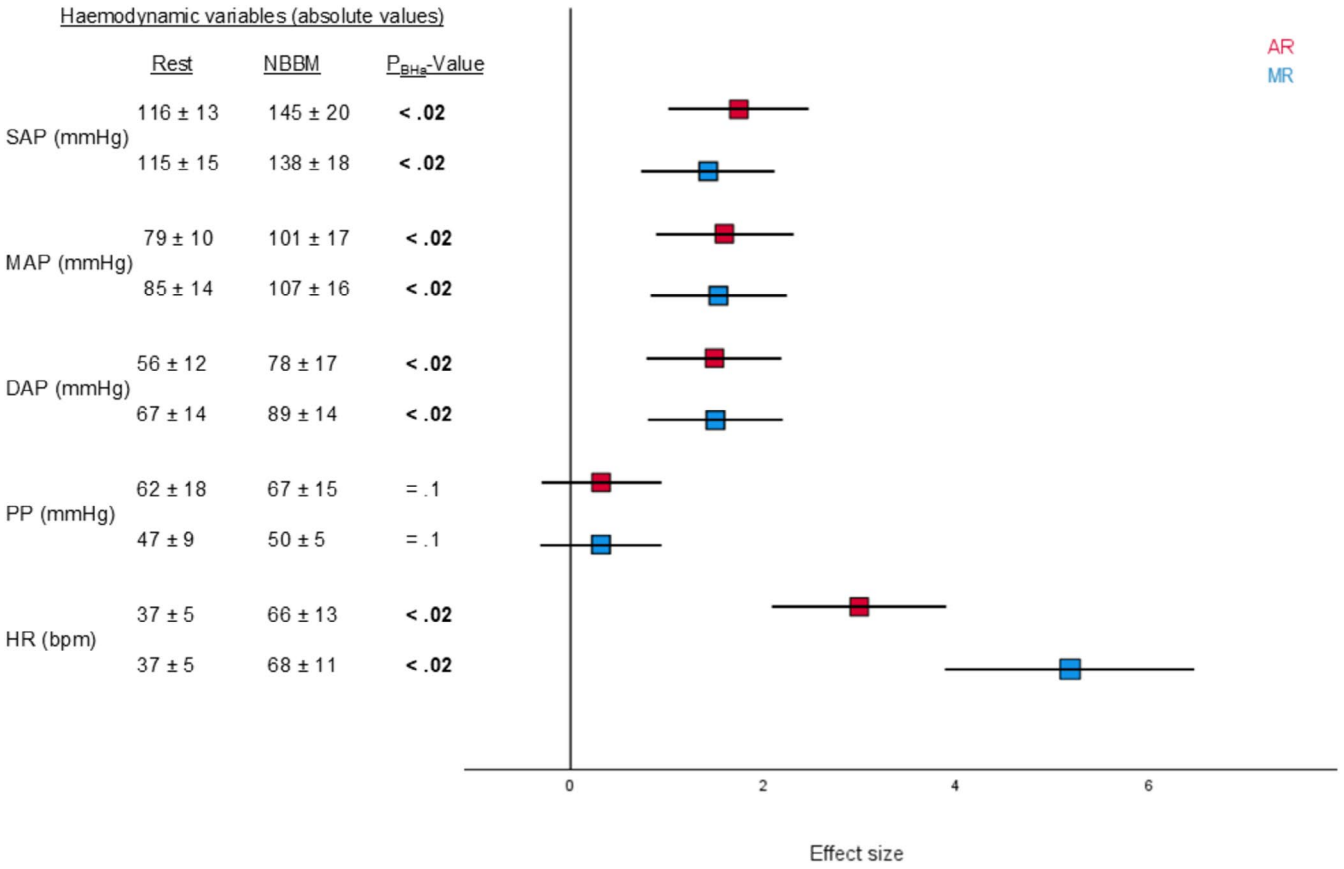

FIGURE 1 | Forest plot of effect sizes of N-butylscopolammonium bromide and metamizol sodium challenge along with absolute values for heart rate, pulse pressure as well as non-corrected diastolic, mean and systolic arterial blood pressures in horses with aortic and mitral valve regurgitation. The x-axis displays the effect size of the pharmacological challenge (box) and its 95% confidence interval of the result (horizontal line). Result lines crossing the vertical line of null effect represent statistically non-significant results. Abbreviations: AR, aortic regurgitation; DAP, diastolic arterial pressure; HR, heart rate; MAP, mean arterial pressure; MR, mitral regurgitation; PP, pulse pressure; SAP, systolic arterial pressure.

Cardiac murmur timing did not differ significantly between pharmacological challenge and rest (Table 3). Detection of the presence of a systolic or diastolic cardiac murmur showed substantial to almost perfect interobserver agreement, whereas agreement was lower for murmur timing and intensity. Although not statistically significant, some mid-to-end systolic murmurs became holosystolic after NBBM administration in horses with MR ($n = 5$ based on live auscultation, $n = 4$ based on recording). Median cardiac murmur intensity based on the recording increased after pharmacological challenge with NBBM compared with rest for horses with AR ($P_{BHa} = 0.03$) and horses with MR ($P_{BHa} < 0.02$; Table 3). The median intensity of the cardiac murmur assessed by three observers increased after administration of NBBM in 14/20 horses with AR and 13/20 horses with MR, whereas the intensity remained the same in 6/20 horses with AR and 7/20 horses with MR. After exercise testing, median cardiac murmur intensity increased in 9/13 horses with AR and 13/15 horses with MR, remained the same in 3/13 horses with AR and 2/15 horses with MR, and decreased in 1/13 horses with AR.

Atrial premature depolarizations were noticed during echocardiography at rest in 4/20 horses with AR and 8/20 horses with MR, during exercise in one horse with AR and 3/15 horses with MR as well as during pharmacological challenge in 6/20 horses with AR and 2/20 with MR. Isolated ventricular premature depolarizations were noticed during echocardiography at rest in 2/20 horses with AR and 2/20 horses with MR, during exercise in one horse with AR and 2/15 horses with MR, and after pharmacological challenge in 2/20 horses with MR. Premature depolarizations were not consistently observed in the same individual horses at rest and during exercise or pharmacological challenge.

4 | Discussion

Intravenous administration of NBBM resulted in significantly increased HR and BP with alterations in cardiac dimensions and functional parameters in horses with AR and MR. Pharmacological challenge resulted in significantly decreased LA areas and LV end-diastolic area, volume, and internal diameter as previously described in healthy Warmblood horses [18]. Inhibition of the muscarinic M2 receptors of the parasympathetic nervous system leads to tachycardia which subsequently results in a decreased atrial and ventricular filling time, and as such in a decreased cardiac preload [16, 29, 30]. Compared with a previous study in healthy horses, the mean differences in echocardiographic parameters before and after administration of NBBM were larger in horses with AR and MR [18]. Valvular regurgitation is characterized by regurgitant blood flow toward the receiving chamber (i.e., the LV in case of AR and the LA in

TABLE 1 | Results of echocardiographic variables in horses with aortic and mitral valve regurgitation at rest and after pharmacological challenge with NBBM.

		AR					MR				
		Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p *F*-test	P_{BHa}	Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p *F*-test	P_{BHa}
2D—Left atrial dimensions and function (R-LVLAX view focused on the LA)											
L-LADs_{end}LAX (500)	cm	11.5 ± 0.8	10.6 ± 0.8	−0.9 (−1.2 to −0.6)	< 0.001	**< 0.02**	11.4 ± 0.9	11.0 ± 0.9*	−0.5 (−0.7 to −0.2)	< 0.001	**< 0.02**
L-LAD$_{max}$s$_{end}$LAX (500)	cm	12.5 ± 0.8	11.7 ± 1.0	−0.8 (−1.1 to −0.5)	< 0.001	**< 0.02**	12.4 ± 0.9	11.7 ± 0.8	−0.7 (−1.0 to −0.4)	< 0.001	**< 0.02**
R-LADs_{end}LAX (500)	cm	10.2 ± 1.0	9.9 ± 0.9	−0.3 (−0.7 to 0.2)	0.001	**0.02**	10.1 ± 0.8	9.5 ± 1.0	−0.6 (−0.9 to −0.3)	0.001	**0.02**
R-LAD$_{max}$s$_{end}$LAX (500)	cm	11.9 ± 1.0	11.2 ± 1.0	−0.7 (−1.0 to −0.3)	< 0.001	**< 0.02**	11.7 ± 1.1	10.4 ± 1.1*	−1.3 (−1.7 to −0.9)	< 0.001	**< 0.02**
R-LAAs_{end}LAX (500)	cm^2	91.5 ± 13.0	76.9 ± 12.2	−14.6 (−18.3 to 10.9)	< 0.001	**< 0.02**	87.4 ± 15.3	74.7 ± 12.8	−12.6 (−18.9 to −6.4)	< 0.001	**< 0.02**
R-LAAd_{end}LAX (500)	cm^2	61.8 ± 11.3	58.6 ± 11.3	−3.2 (−8.0 to 1.6)	< 0.001	**< 0.02**	61.5 ± 10.4	53.7 ± 8.8	−7.9 (−11.9 to −3.9)	< 0.001	**< 0.02**
R-LAApLAX (500)	cm^2	77.6 ± 12.5	72.0 ± 13.7	−5.6 (−8.4 to −2.9)	< 0.001	**< 0.02**	77.5 ± 11.2	63.3 ± 11.2*	−14.3 (−18.0 to −10.5)	< 0.001	**< 0.02**
R-LAAaLAX (500)	cm^2	59.6 ± 10.5	49.0 ± 9.6	−10.7 (−13.5 to −7.8)	< 0.001	**< 0.02**	58.9 ± 9.0	43.9 ± 8.1	−15.0 (−18.6 to −11.5)	< 0.001	**< 0.02**
LA FAC LAX (500)	%	22.9 ± 8.3	31.6 ± 7.6	8.7 (6.0 to 11.4)	< 0.001	**< 0.02**	23.9 ± 5.7	30.4 ± 7.6	6.5 (2.5 to 10.5)	< 0.001	**< 0.02**
2D—Left ventricular dimensions and function (R-LVLAX view focused on the LV)											
R-LVAd_{end}LAX (500)	cm^2	181.3 ± 31.2	154.3 ± 34.3	−27.0 (−36.0 to −18.0)	< 0.001	**< 0.02**	149.4 ± 16.6	129.0 ± 14.7	−20.4 (−27.5 to −13.3)	< 0.001	**< 0.02**
R-LVAs_{end}LAX (500)	cm^2	75.3 ± 18.3	76.4 ± 18.6	1.1 (−3.2 to 5.4)	0.2	0.3	64.1 ± 11.1	67.5 ± 12.2	3.4 (−1.6 to 8.3)	0.2	0.3
LV FAC LAX (500)	%	58.8 ± 4.9	50.5 ± 5.5	−8.3 (−10.1 to −6.5)	< 0.001	**< 0.02**	57.2 ± 4.0	47.8 ± 6.3	−9.4 (−11.7 to −7.1)	< 0.001	**< 0.02**
LVV$d_{end,ModSi}$ (500)	mL	1573 ± 442	1228 ± 448	−345 (−459 to −231)	< 0.001	**< 0.02**	1139 ± 197	918 ± 172	−220 (−296 to −145)	< 0.001	**< 0.02**
LVV$s_{end,ModSi}$ (500)	mL	406 ± 145	420 ± 140	15 (−23 to 53)	0.4	0.5	322 ± 84	328 ± 96	6 (−29 to 41)	0.4	0.5

(Continues)

TABLE 1 | (Continued)

		AR					MR				
		Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p *F*-test	P_{BHa}	Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p *F*-test	P_{BHa}
LV SV_{ModSi} (500)	mL	1167 ± 316	807 ± 335	−360 (−453 to −267)	< 0.001	**< 0.02**	817 ± 132	590 ± 114*	−226 (−277 to −175)	< 0.001	**< 0.02**
LV EF_{ModSi} (500)	%	74.5 ± 4.3	65.1 ± 5.7	−9.4 (−11.6 to −7.2)	< 0.001	**< 0.02**	71.9 ± 3.9	64.5 ± 6.7	−7.4 (−9.6 to −5.0)	< 0.001	**< 0.02**
LV CO_{ModSi} (500)	L/min	43.5 ± 13.6	54.5 ± 15.9	11.0 (5.6 to 16.4)	< 0.001	**< 0.02**	30.1 ± 6.7	43.4 ± 8.6	13.4 (9.9 to 16.9)	< 0.001	**< 0.02**
2D—Aortic and pulmonary variables (R-LVOT view)											
R-LVOT $AoDd_{end}$ (500)	cm	6.1 ± 0.5	6.8 ± 0.6	0.7 (0.5 to 0.9)	< 0.001	**< 0.02**	5.9 ± 0.5	6.6 ± 0.5	0.6 (0.5 to 0.8)	< 0.001	**< 0.02**
R-LVOT $AoDs_{peak}$ (500)	cm	6.7 ± 0.5	7.1 ± 0.5	0.4 (0.2 to 0.6)	< 0.001	**< 0.02**	6.4 ± 0.6	6.9 ± 0.5	0.5 (0.3 to 0.6)	< 0.001	**< 0.02**
R-$LVOTs_{peak}$ (500)	cm	6.2 ± 0.7	6.0 ± 0.8	−0.2 (−0.4 to 0.1)	0.04	0.1	5.9 ± 0.6	5.8 ± 0.6	−0.2 (−0.4 to 0.1)	0.04	0.1
R-LVOT $PADd_{end}$ (500)	cm	4.8 ± 0.3	5.6 ± 0.4	0.8 (0.6 to 1.0)	< 0.001	**< 0.02**	4.8 ± 0.5	5.4 ± 0.5	0.6 (0.4 to 0.8)	< 0.001	**< 0.02**
R-LVOT $PADs_{peak}$ (500)	cm	5.4 ± 0.6	5.8 ± 0.3	0.4 (0.1 to 0.7)	< 0.001	**< 0.02**	5.2 ± 0.4	5.9 ± 0.5	0.7 (0.5 to 0.9)	< 0.001	**< 0.02**
2D short-axis variables (R-AoSAX view)											
R-$AoDs_{end}$SAX (500)	cm	8.1 ± 0.7	8.1 ± 0.6	−0.0 (−0.3 to 0.2)	0.2	0.3	7.6 ± 0.5	7.8 ± 0.7	0.2 (0.0 to 0.4)	0.2	0.3
R-$LADs_{end}$SAX (500)	cm	9.3 ± 0.5	9.4 ± 0.7	0.0 (−0.3 to 0.4)	0.8	0.9	9.5 ± 0.5	9.1 ± 0.8*	−0.4 (−0.6 to −0.1)	0.01	0.08
R-$LADs_{end}$SAX/$AoDs_{end}$SAX (500)		1.2 ± 0.1	1.2 ± 0.1	0.0 (−0.0 to 0.1)	0.8	0.9	1.3 ± 0.1	1.2 ± 0.1*	−0.1 (−0.1 to 0.0)	< 0.001	**< 0.02**
M-mode—Ventricular dimensions (R-LV_{ch}SAX view)											
$RVIDd_{end}$ (500)	cm	2.8 ± 0.8	2.7 ± 0.9	−0.1 (−0.4 to 0.2)	0.01	0.08	3.7 ± 0.6	3.2 ± 0.8	−0.5 (−0.9 to −0.2)	0.012	0.08

(Continues)

TABLE 1 | (Continued)

		AR					MR				
		Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p *F*-test	P_{BHa}	Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p *F*-test	P_{BHa}
$RVIDs_{peak}$ (500)	cm	2.3 ± 0.7	1.7 ± 0.8	−0.6 (−1.0 to −0.3)	< 0.001	**< 0.02**	2.9 ± 0.7	2.0 ± 0.7	−0.9 (−1.3 to-0.5)	< 0.001	**< 0.02**
$IVSd_{end}$ (500)	cm	3.0 ± 0.3	3.8 ± 0.4	0.8 (0.5 to 0.9)	< 0.001	**< 0.02**	3.2 ± 0.4	3.5 ± 0.4*	0.3 (0.2 to 0.5)	0.001	**0.02**
$IVSs_{peak}$ (500)	cm	4.3 ± 0.5	4.2 ± 0.4	−0.0 (−0.3 to 0.2)	1.0	1.0	4.1 ± 0.4	4.1 ± 0.5	0.0 (−0.3 to 0.3)	1.0	1.0
$LVIDd_{end}$ (500)	cm	12.7 ± 1.6	11.9 ± 2.0	−0.8 (−1.2 to −0.4)	< 0.001	**< 0.02**	11.2 ± 0.9	10.9 ± 1.1*	−0.3 (−0.5 to 0.0)	0.04	0.09
$LVIDs_{peak}$ (500)	cm	7.9 ± 1.1	7.8 ± 1.5	−0.1 (−0.7 to 0.5)	0.8	0.9	6.8 ± 1.0	6.8 ± 0.8	−0.0 (−0.4 to 0.3)	0.8	0.9
$LVFWd_{en}$ (500)	cm	2.5 ± 0.3	2.7 ± 0.3	0.2 (0.0 to 0.4)	0.003	**0.03**	2.4 ± 0.3	2.5 ± 0.4	0.2 (0.0 to 0.3)	0.003	**0.03**
$LVFWs_{peak}$ (500)	cm	4.0 ± 0.4	4.2 ± 0.5	0.2 (0.0 to 0.4)	0.09	0.2	4.2 ± 0.5	4.2 ± 0.5	0.0 (−0.2 to 0.2)	0.09	0.2
LV FS (500)	%	37.9 ± 4.5	34.0 ± 7.0	−4.0 (−8.0 to 0.1)	0.03	0.09	39.1 ± 5.5	37.5 ± 5.8	−1.6 (−4.5 to 1.4)	0.03	0.09
$RWTd_{end}$ (500)		0.4 ± 0.1	0.6 ± 0.1	0.1 (0.1 to 0.2)	< 0.001	**< 0.02**	0.5 ± 0.1	0.6 ± 0.1*	0.1 (0.0 to 0.1)	< 0.001	**< 0.02**
PWD—Aortic flow (L-LVOT view)											
L-PWD AoV Vel_{max}	m/s	1.0 ± 0.2	1.0 ± 0.2	−0.1 (−0.1 to 0.0)	0.02	0.09	0.8 ± 0.1	1.0 ± 0.2*	0.1 (0.1 to 0.2)	0.003	**0.03**
L-PWD AoV VTI	cm	45.3 ± 9.2	29.4 ± 5.5	−15.8 (−19.8 to −11.9)	< 0.001	**< 0.02**	32.8 ± 5.3	28.8 ± 5.1*	−4.0 (−7.7 to −0.3)	0.04	0.09
L-PWD AoV PEP	ms	78 ± 15	122 ± 26	44 (34 to 54)	< 0.001	**< 0.02**	105 ± 18	141 ± 26	37 (24 to 49)	< 0.001	**< 0.02**
L-PWD AoV ET	ms	585 ± 141	404 ± 85	−181 (−244 to −118)	< 0.001	**< 0.02**	557 ± 79	398 ± 90	−160 (−213 to −107)	< 0.001	**< 0.02**
L-PWD AoV PEP/ET		0.1 ± 0.0	0.3 ± 0.1	0.2 (0.1 to 0.2)	< 0.001	**< 0.02**	0.2 ± 0.0	0.4 ± 0.1	0.2 (0.1 to 0.2)	< 0.001	**< 0.02**
L-PWD AoV SV (500)	mL	1328 ± 465	836 ± 333	−492 (−627 to −357)	< 0.001	**< 0.02**	1126 ± 290	901 ± 569	−224 (−491 to 43)	< 0.001	**< 0.02**

(Continues)

TABLE 1 | (Continued)

		AR					MR				
		Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p F-test	P_{BHa}	Rest ($N=20$)	NBBM ($N=20$)	Mean difference (95% CI)	p F-test	P_{BHa}
L-PWD AoV CO (500)	L/min	48.5 ± 16.3	53.4 ± 21.1	4.9 (−3.5 to 13.2)	0.02	0.09	41.4 ± 15.3	57.9 ± 35.9	16.4 (−0.4 to 33.3)	0.02	0.09
PW tissue Doppler (R-LV$_{ch}$SAX view)											
E_m	cm/s	31 ± 8	32 ± 8	1 (−4 to 6)	1.0	1.0	33 ± 5	31 ± 6	−1 (−4 to 1)	1.0	1.0
A_m	cm/s	12 ± 4	15 ± 8	3 (0 to 6)	0.03	0.09	12 ± 5	13 ± 4	1 (−1 to 3)	0.03	0.09
E_m/A_m		2.8 ± 1.0	3.2 ± 2.4	0.4 (−0.6 to 1.4)	0.9	1.0	3.3 ± 1.6	2.9 ± 1.3	−0.5 (−1.3 to 0.4)	0.9	1.0
S_m	cm/s	13 ± 2	13 ± 2	0.1 (−0.9 to 1.0)	0.5	0.6	13 ± 1	12 ± 2	−0.4 (−1.3 to 0.4)	0.5	0.6
Regurgitant jet area											
R-Regurgitant jet area	cm^2	17.9 (11.3; 29.7)	31.2 (16.7; 41.2)	/	< 0.001	**< 0.02**	/	/	/	/	/
L-Regurgitant jet area	cm^2	17.9 (10.0; 28.0)	19.9 (13.4; 37.9)	/	0.001	**0.02**	14.3 (6.7; 19.9)	21.7 (13.7; 31.4)	/	0.002	**0.04**

Note: Mean values and standard deviation (SD) are reported for normally distributed data, median and interquartile range (Q1; Q3) are reported for non-normally distributed data. Mean difference and its 95% confidence interval (CI) are reported for normally distributed data. Normally distributed data during rest and NBBM challenge were compared by general linear model analysis for repeated measures. Variables highlighted by an asterisk (*) indicate variables where pharmacological challenge resulted in a different treatment effect in horses with MR compared to horses with AR. If pharmacological challenge resulted in a significant different treatment effect ($p < 0.05$), variables were analyzed by a paired sample t-test and its respective p-value is indicated. If data were not normally distributed, a related samples Wilcoxon signed rank test was performed to compare medians before and after NBBM challenge. Benjamini-Hochberg adjusted p-values (P_{BHa}) are reported along with the original p-values and P_{BHa} values < 0.05 were considered significant (shown in bold). Diameters, areas, volumes and calculated variables were normalized to a BW of 500 kg using the formulae of allometric scaling.

Abbreviations: A_m, peak LV radial wall motion velocity during late diastole; Ao, Aortic; AoDd$_{end}$LAX, Ao diameter at sinotubular junction at end-diastole on LAX; AoDs$_{end}$SAX, Ao diameter at end-systole on SAX; AoDs$_{peak}$LAX, Ao diameter at sinotubular junction at peak-systole on LAX; AoV, aortic valve; AR, aortic regurgitation; bpm, beats per minute; ch, chordal level; DAP, diastolic arterial pressure; E_m, peak LV radial wall motion velocity during early diastole; E_m/A_m, ratio of peak LV radial wall motion velocity during early and late diastole; HR, heart rate; IVSd$_{end}$, interventricular septal thickness at end-diastole at chordal level determined by M-mode; IVSs$_{peak}$, interventricular septal thickness at peak-systole at chordal level determined by M-mode; L, left-sided; LA FAC LAX, LA fractional area change; LA, left atrium; LAAaLAX, LA area at maximal atrial contraction on a R-LVLAX view; LAAd$_{end}$LAX, LA area at end-diastole on a R-LVLAX view; LAApLAX, LA area at the onset of the P wave on a R-LVLAX view; LAAs$_{end}$LAX, LA area at end-systole on a R-LVLAX view; LAD$_{max}$s$_{end}$LAX, LA diameter at end-systole on a LVLAX view; LADs$_{end}$LAX, LA diameter at end-systole on a LVLAX view; LADs$_{end}$SAX, LA diameter at end-systole on SAX; LADs$_{end}$SAX/AoDs$_{end}$SAX, ratio of the LA and Ao short axis diameter at end-systole; LAX, long axis; LV CO$_{ModSi}$LAX, LV cardiac output determined by Simpson's modified method of disks on LAX; LV EF$_{ModSi}$LAX, LV ejection fraction determined by Simpson's modified method of disks on LAX; LV FAC LAX, LV fractional area change; LV FS, LV fractional shortening; LV SV$_{ModSi}$LAX, LV stroke volume determined by Simpson's modified method of disks on LAX; LV, left ventricle; LVAd$_{end}$LAX, LV area at end-diastole on LAX; LVAs$_{end}$LAX, LV area at end-systole on LAX; LVFWd$_{end}$, LV free wall thickness at end-diastole at chordal level determined by M-mode; LVFWs$_{peak}$, LV free wall thickness at peak-systole at chordal level determined by M-mode; LVIDd$_{end}$, LV internal diameter at end-diastole at chordal level determined by M-mode; LVIDs$_{peak}$, LV internal diameter at peak-systole at chordal level determined by M-mode; LVOTs$_{peak}$LAX, Left ventricular outflow tract diameter at peak-systole on LAX; LVVd$_{end,ModSi}$LAX, LV volume at end-diastole determined by Simpson's modified method of disks on LAX; LVVs$_{end,ModSi}$LAX, LV volume at end-systole determined by Simpson's modified method of disks on LAX; MAP, mean arterial pressure; ModSi, modified Simpson's method of disks; MR, mitral regurgitation; NBBM, N-butylscopolammonium bromide and metamizol sodium; PADd$_{end}$SAX, PA diameter at end-diastole on SAX; PADs$_{peak}$SAX, PA diameter at peak-systole on SAX; PP, pulse pressure; PW, Pulsed Wave; PWD, Pulsed wave Doppler; PWD AoV CO, LV cardiac output across the aortic valve assessed by PWD; PWD AoV ET, ejection time assessed by PWD; PWD AoV PEP, pre-ejection period assessed by PWD; PWD AoV PEP/ET, ratio of pre-ejection period and ejection time; PWD AoV SV, stroke volume across the aortic valve assessed by PWD; PWD AoV Vel$_{max}$, maximal aortic flow velocity across the aortic valve during systole assessed by PWD; PWD AoV VTI, velocity time integral assessed by PWD; R, right-sided; RVIDd$_{end}$, RV internal diameter at end-diastole at chordal level determined by M-mode; RVIDs$_{peak}$, RV internal diameter at peak-systole at chordal level determined by M-mode; RWTd$_{end}$, LV relative wall thickness at end-diastole determined at chordal level by M-mode; SAP, systolic arterial pressure; SAX, short axis; S_m, peak LV radial wall motion velocity during peak-systole.

TABLE 2 | Results of regurgitation jet score and jet size after pharmacological challenge with N-butylscopolammonium bromide and metamizol sodium (NBBM) compared with rest.

	AR				MR			
			p				*p*	
	Rest (*N* = 20)	NBBM (*N* = 20)	*F*-test	P_{BHa}	Rest (*N* = 20)	NBBM (*N* = 20)	*F*-test	P_{BHa}
Jet score								
Score 0	1	1			11	4		
Score 1	7	6			8	8		
Score 2	6	3			1	5		
Score 3	6	10			0	3		
Median score	2 (1; 3)	3 (1; 3)	0.03	0.09	0 (0; 1)	1 (1; 2)	0.003	**0.03**
Jet score compared to rest								
Equal score		*n* = 15				*n* = 9		
Increased score		*n* = 5				*n* = 11		
Subjective appearance of jet size compared to rest								
Similar score		*n* = 3				*n* = 4		
Increased		*n* = 17				*n* = 16		

Note: Benjamini-Hochberg adjusted *p*-values (P_{BHa}) are reported along with the original *p*-values and P_{BHa} values < 0.05 were considered significant (shown in bold).

case of MR). Tachycardia results in a shortened time for regurgitant blood flow toward the receiving chamber. This factor might explain the larger mean differences in diastolic LV dimensions and systolic LA dimensions compared with healthy horses undergoing the same drug effects. Left ventricular dimensions in healthy horses were determined using the area-length method and not corrected for body weight, whereas our study used the modified Simpson's method of disks and allometric scaling [18]. However, a previous study demonstrated that the area-length method and the modified Simpson's method of disks result in very similar LV estimates [31]. Additionally, a previous study on cardiac output measurements using different echocardiographic methods showed no significant difference between both methods [32]. Allometric scaling, on the other hand, influences the absolute measurement values, especially those of diastolic LV dimensions and cardiac output (CO) [31], but should not have influenced relative differences between means before and after pharmacological challenge, allowing comparison of the data of both studies. A decrease in LV end-diastolic internal diameter after NBBM administration was noticed in horses with AR, whereas this echocardiographic variable remained unchanged in horses with MR. The duration of the diastolic filling phase within the cardiac cycle determines end-diastolic atrial and ventricular dimensions, as mentioned above. In horses with AR, a longer diastole additionally leads to an increased regurgitant blood volume from the aorta toward the LV, resulting in progressive LV dilatation throughout diastole [33, 34]. The shortened diastolic filling time arising from tachycardia leads to horses with AR having decreased regurgitant blood flow toward the LV, which results in a larger decrease in LV diastolic dimensions. Left ventricular systolic variables were not altered significantly both in AR and MR, suggesting that LV systolic dimensions are less prone to NBBM drug effects (Table 1). A relative decrease in LA systolic variables would be expected in the case of MR but was not identified in our study. This finding might be explained by the fact that the overall severity grade of MR was lower compared with the severity grade of AR. Inclusion of horses with a higher severity grade of MR might have resulted in accentuated drug effects on systolic LA dimensions.

Pharmacological challenge with NBBM led to an increased subjectively assessed jet size and to an increased regurgitant jet area in horses with AR and MR as well as to an increased jet score in horses with MR (Tables 1 and 2). Administration of NBBM resulted in higher BP, which leads to increased sinotubular aortic and pulmonary dimensions [18, 35], and which might also explain the accentuated regurgitant jet. The parasympatholytic actions of NBB are thought to be the causative factor for the observed increases in BP because catecholamine concentrations have been reported to be unaffected after pharmacological challenge with NBB in humans [16, 29]. Although the subjectively assessed regurgitant jet size appeared enlarged in the majority of horses, increased jet scores were only assigned in 5/20 horses with AR and 11/20 horses with MR. This discrepancy between jet size and jet score can be explained by the fact that the intensity of the regurgitant jet might be increased while being categorized under the same jet score (Figure 2). These findings emphasize that HR and BP have an important effect on cardiac dimensions, function, and regurgitant blood flow in horses with valvular regurgitation. Higher HR might result in underestimation of the hemodynamic impact of valvular regurgitation because of decreased chamber sizes. On the other hand, jet size might be overestimated as a result of higher BP. The clinician should be aware of these influencing factors when determining the clinical impact of valvular regurgitation on performance, life expectancy, and rider safety. In addition, when performing long-term follow-up, one should realize that differences in HR and BP affect cardiac dimensions and regurgitant jet size.

TABLE 3 | Results of cardiac auscultation. Summary of the auscultation data of the left hemithorax for the live auscultation and the recording of the cardiac murmurs assessed by three independent observers.

	Aortic regurgitation (diastolic murmur)							Mitral regurgitation (systolic murmur)						
	Rest ($N=20$)	**Exercise ($N=13$)**	***p* *F*-test**	**P_{BHa}**	**NBBM ($N=20$)**	***p* *F*-test**	**P_{BHa}**	**Rest ($N=20$)**	**Exercise ($N=15$)**	***p* *F*-test**	**P_{BHa}**	**NBBM ($N=20$)**	***p* *F*-test**	**P_{BHa}**
Cardiac murmur timing														
Live auscultation			0.3	0.5		1.0	1.0			0.07	0.1		0.2	0.3
No murmur	$n=2$	$n=0$			$n=2$			$n=2$	$n=2$			$n=2$		
Early-mid	$n=0$	$n=0$			$n=0$			$n=0$	$n=1$			$n=0$		
Mid-end	$n=1$	$n=0$			$n=1$			$n=16$	$n=6$			$n=11$		
Holo	$n=17$	$n=13$			$n=17$			$n=2$	$n=6$			$n=7$		
Pan	$n=0$	$n=0$			$n=0$			$n=0$	$n=0$			$n=0$		
Recording			0.1	0.2		0.2	0.3			0.5	0.6		0.4	0.5
No murmur	$n=5$	$n=0$			$n=2$			$n=1$	$n=0$			$n=1$		
Early-mid	$n=1$	$n=0$			$n=0$			$n=1$	$n=1$			$n=1$		
Mid-end	$n=0$	$n=0$			$n=0$			$n=8$	$n=3$			$n=4$		
Holo	$n=12$	$n=11$			$n=17$			$n=6$	$n=7$			$n=12$		
Pan	$n=0$	$n=0$			$n=0$			$n=0$	$n=0$			$n=0$		
Cardiac murmur intensity														
Median of live auscultation	3 (2; 4)	4 (3; 4)	0.08	0.1	4 (3; 4)	0.03	0.09	3 (2; 4)	4 (2; 4)	0.06	0.1	4 (2; 4)	0.03	0.09
Median of recording (3 observers)	2 (0; 3)	3 (2; 3)	0.02	0.09	3 (2; 4)	0.003	**0.03**	2 (1; 3)	3 (2; 4)	0.002	0.03	3 (2; 4)	<0.001	**<0.02**

Note: Murmur timing based on the recordings was reported if at least 2 of 3 observers agreed on the classification. Cardiac murmur timing (no cardiac murmur, early-mid, mid-end, holo, pan) and median (Q1; Q3) intensity (grade 0–5/6) of the cardiac murmur were compared between rest and exercise and between rest and administration of N-butylscopolammonium bromide and metamizol sodium (NBBM). Benjamini-Hochberg adjusted *p*-values (P_{BHa}) are reported along with the original *p*-values and P_{BHa} values <0.05 were considered significant (shown in bold).

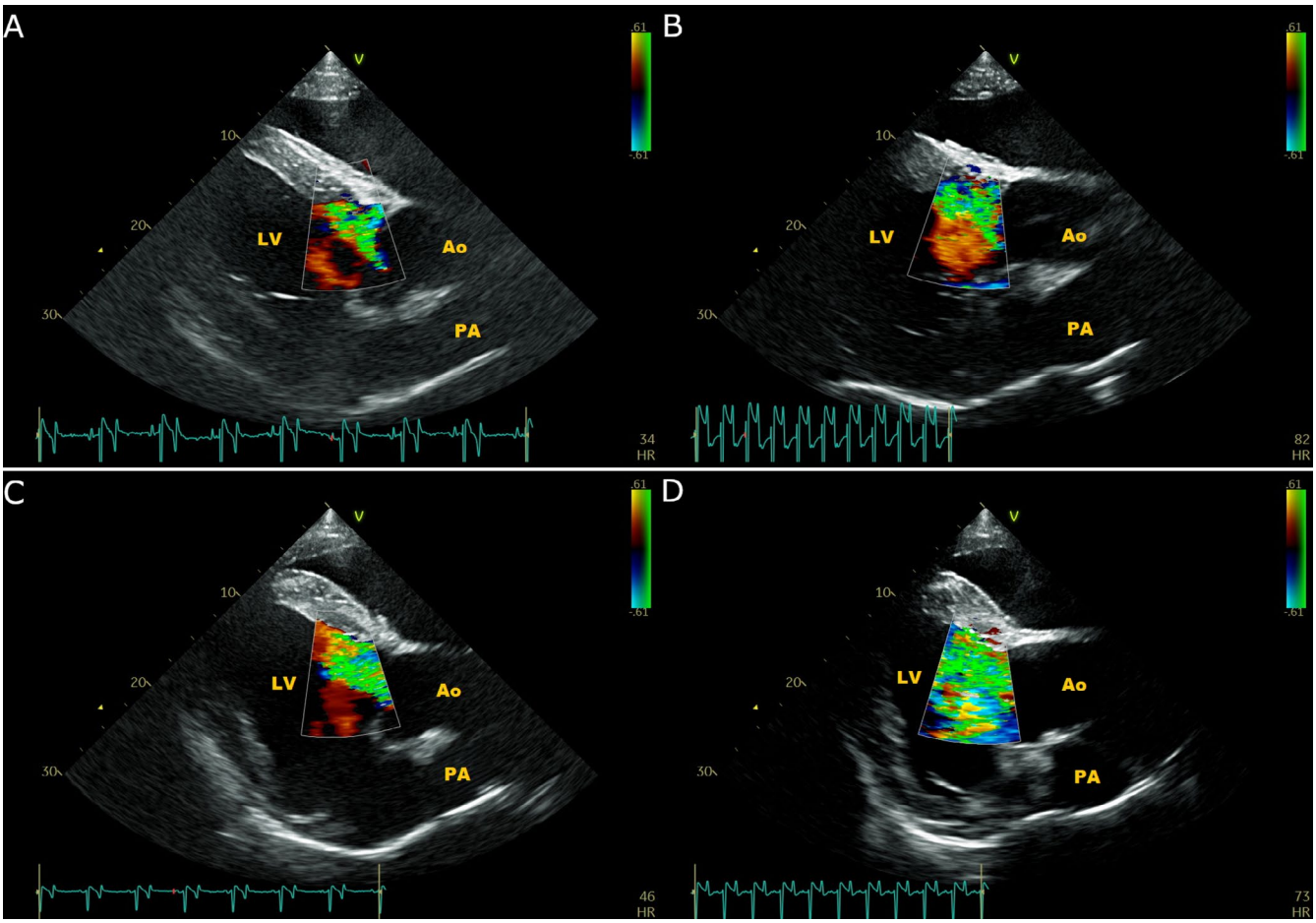

FIGURE 2 | Evaluation of aortic regurgitation in two horses during pharmacological challenge with N-butylscopolammonium bromide and metamizol sodium (NBBM). (A and B) show a right parasternal left ventricular outflow tract (R-LVOT) view of a 17-year old Warmblood mare with a jet score of 1: NBBM administration (B) resulted in the same jet score, although visually a minimally increased jet size compared to rest (A) was present. (C and D) show a R-LVOT view of a 20-year old Warmblood mare: Compared to rest (C, jet score 1), NBBM administration resulted in a visually increased jet size and jet score (D, jet score 3). Abbreviations: Ao, aorta; LV, left ventricle; PA, pulmonary artery.

Pharmacological challenge with NBBM led to an increase in cardiac murmur intensity in horses with AR and MR, probably as a result of higher BP. Cardiac auscultation has been described as a diagnostic technique to detect cardiac disease, but neither timing nor grade of the murmur accurately predict severity or prognosis for athletic performance and longevity [3]. In our study, exercise testing resulted in a similar or increased cardiac murmur intensity compared to rest. This observation is in contrast with previous findings that noted that a majority of small mitral and aortic valve regurgitations disappeared or decreased upon auscultation or echocardiographic assessment after light exercise testing compared with rest [7]. However, in this study on Standardbred trotters, tricuspid valve insufficiency intensified during auscultation and echocardiography after exercise testing in horses with clinical signs of poor performance [7]. Median intensity of cardiac murmurs increased in our study in 65% of the examined horses after administration of NBBM. This observation is consistent with anecdotal reports that administration of NBBM in horses with colic accentuates cardiac murmurs by anticholinergic drug effects [21]. Whether murmurs in horses with colic are caused by ejection or by valvular regurgitation, which are softer or inaudible under resting conditions, remains to be determined. However, it is important to consider altered auscultatory findings caused by differences in HR and BP. Therefore, cardiac auscultation both at physiological HRs in absence of external stressors and after stress testing might be of clinical utility for the severity assessment of horses with valvular regurgitation.

The aim of our trial was to evaluate alterations in valvular regurgitation as a result of pharmacological challenge with NBBM in horses. Variability in individual responses might be useful for a more comprehensive evaluation of horses with valvular regurgitation. The use of NBBM might help distinguish horses at higher risk of disease progression from those with a more stable condition [7, 36]. In human medicine, stress testing in patients with valvular regurgitation provides valuable diagnostic insights, particularly in detecting volume or pressure overload in the pulmonary circulation and identifying LV systolic dysfunction that may not be evident at rest [9, 37, 38]. Pharmacological stress testing using dobutamine or adenosine in human medicine primarily aims to assess coronary artery perfusion and myocardial wall movement, but also is used to assess regurgitant and transvalvular flow [9, 37, 39]. Our findings showed that NBBM administration enhanced the visibility of the regurgitant jets (i.e., jet size). At this time, it remains unclear whether an

increase in jet size after NBBM administration indicates a more pronounced negative effect on cardiac function during exercise or more rapid disease progression. Larger studies with long-term follow-up are needed to determine the diagnostic value of pharmacological challenge using NBBM.

Our study had several limitations. Overall valvular insufficiency grade was more severe in the AR compared with the MR group. Differences in severity grade might have influenced the results of our study, especially those on LA systolic dimensions in horses with MR. Concurrent valvular regurgitation might have influenced end-diastolic left ventricular dimensions in cases of concurrent AR, or end-systolic atrial dimensions in cases of concurrent MR, when compared with horses without additional regurgitation. However, these effects were considered minimal because the impact of concurrent regurgitation was estimated to be less significant than that of the primary one. Regurgitant area measurement on parasternal views generally is not recommended for the severity assessment of valvular regurgitation in adult horses, but this method was chosen in an attempt to objectively quantify the effect of NBBM [2]. Blood pressure was measured noninvasively and not corrected to the level of the heart base, which might be prone to error compared to invasively measured blood pressure measurements. However, the main goal was to evaluate differences in BP between rest and pharmacological challenge [25]. The order in which image acquisition occurred during pharmacological challenge might have influenced the results because images acquired shortly after drug administration might be more prone to drug effects. To counteract this effect, image acquisition was randomly initiated either on the left or right side of the horse. Exercise testing might have influenced the hydration state of the horses, and neither total protein concentration nor packed cell volume was determined before pharmacological challenge. However, 1 h of rest before pharmacological challenge with access to fresh water should have allowed compensation for the fluid losses of the exercise test. The recorded auscultation files or echocardiographic images were analyzed by observers who were blinded to the respective horse, but not to treatment because the increased HR was obvious. Volumetric and PWD-derived calculated echocardiographic variables are based on geometric assumptions and consequently are prone to error. Variables derived from PWD calculations especially rely on optimal alignment and cursor positioning. For this reason, echocardiography and offline analysis were performed by the same operator to minimize interobserver variability [2]. Interobserver variability in grading of cardiac murmur timing or intensity might have influenced the results of our study. Median cardiac murmur timing was reported when at least 2 of 3 observers agreed on classification, which resulted in a lower reported sample size and subsequently decreased statistical power. Furthermore, the quality of the cardiac murmur recording might have influenced its retrospective offline analysis, although recordings were performed using a high-quality stethoscope suitable for such a purpose.

In conclusion, in our exploratory study, pharmacological challenge with NBBM resulted in pronounced effects on cardiac dimensions in horses with AR and MR. Administration of NBBM resulted in increased jet size and jet area, which were associated with increased cardiac murmur intensity. These findings emphasize that HR and BP have important effects on cardiac chamber size and function, the appearance of regurgitant blood flow as well as murmur intensity. Clinicians should take these effects into consideration, especially during follow-up examinations to evaluate the progression of valvular regurgitation. The use of NBBM as a pharmacological challenge test in horses with valvular regurgitation warrants further research to evaluate its diagnostic utility.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved and followed the guidelines of the Ethical Committee (EC) of the Faculty of Veterinary Medicine, Ghent University, Belgium (EC 2020/100). Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Scoring system for aortic valve regurgitation in horses. **Data S2:** Scoring system for mitral valve regurgitation in horses. **Data S3:** Overview of cardiac measurements and imaging planes.